

Midterm Exam

Version A

Friday, October 9, 2026 · in class

Name: _____ Student ID: _____

How this exam works

- **The scenario.** You are the analyst for a seaside café. The owner wants to forecast tomorrow's **gelato units** so the café orders the right amount of stock. You will choose variables, check them, compare models built with LINEST, test whether a model forecasts days it has not seen, and advise the owner.
- **Reason like a modeller.** "All models are wrong, but some are useful." There is no single correct model. You are marked on whether your choices are sensible and **defended with numbers**, not on guessing the model I have in mind.
- **Stage 1 comes first.** It asks what you expect **before** any data. Finish it before you turn the page and do not change it afterwards; later questions ask you to look back at it.
- **Time and points.** 50 points. The suggested time for each stage is printed in its banner; together they come to about 55 minutes.
- **Show your work.** Partial credit is given for correct steps, even if a final answer is wrong. Round to 2 decimal places unless told otherwise. A reference sheet is on the last page.
- **Allowed:** pencil or pen, eraser, ruler, calculator (a phone may be used as a calculator only). **Not allowed:** notes, books, any other use of a phone or other devices, talking.

Stage	What it covers	Points	Score
1	Think first: choose and defend your variables	9	
2	Check each variable on its own	7	
3	Check how the variables relate to each other	7	
4	Build and compare models with LINEST	13	
5	Does the model forecast unseen days?	7	
6	Advise the owner and extend the model	7	
Total		50	

About the data

The café recorded sales and weather on **168 consecutive trading days** (7 April–21 September 2025). The data are **synthetic teaching data**, simulated for this course, not real café records. The outcome you want to forecast is **gelato_units**: the number of two-scoop gelato cups sold that day. Assume the café can get a reliable weather forecast and the beach-event calendar the evening before, and that it sets its own prices.

Table 1. Candidate variables you may use to forecast the outcome.

Variable	What it measures
temperature_c	Daily mean outdoor temperature (°C) during opening hours
sunshine_hours	Hours of sunshine during opening hours
rainfall_mm	Rain (mm) during opening hours; 0 means a dry day
wind_kmh	Mean wind speed (km/h) during opening hours
humidity_pct	Mean relative humidity (%) during opening hours
beach_event	1 if an organised public event is held on the nearby beach, 0 otherwise
weekend	1 if Saturday or Sunday, 0 otherwise (derive from day_of_week)
gelato_price_eur	Posted price (€) of one standard two-scoop cup. Set by the café; on random days it tries a price $\pm$ €0.30 from the usual one
post_change	0 before 30 June, 1 from 30 June (the day the café's new price policy began; derive from date)
iced_coffee_units	Number of iced coffees sold that day
gelato_revenue_eur	gelato_price_eur $\times$ gelato_units (derive)

STAGE 1 · Think first: choose and defend your variables

≈ 8 min · 9 pts

Research question. Which variables from the table on page 1 should the café use to forecast tomorrow's **gelato_units**? Answer using only the table and your knowledge of how a café works: you have not seen any data yet.

- 1 Choose your model.** Pick **four** variables to include. For each, write the **expected sign** (+ or –) of its effect on gelato units and a short **reason** that explains *how* it would change sales. **4 pts**

Variable	Expected sign	Reason (a few words)

- 2 Leave some out.** Name **two** variables from the table that you would **deliberately not use** in a forecasting model, and give one reason for each. Your reasons must come from how the café works (think about *when* each number becomes known and *what causes it*), not from correlations, because you have not seen any. **2 pts**

- 3 What would prove you wrong?** Choose one of your four variables. Describe a LINEST result (a coefficient and/or its t-statistic) that would show your expectation was wrong. **1 pt**

- 4 Data quality.** beach_event is 1 on any day with an organised public event on the beach and 0 otherwise. **2 pts**
 (a) Describe one **validity** problem: how does this coding fail to capture the idea of a crowd-drawing event?
 (b) Describe one **reliability** problem: how could two people record the same day differently?

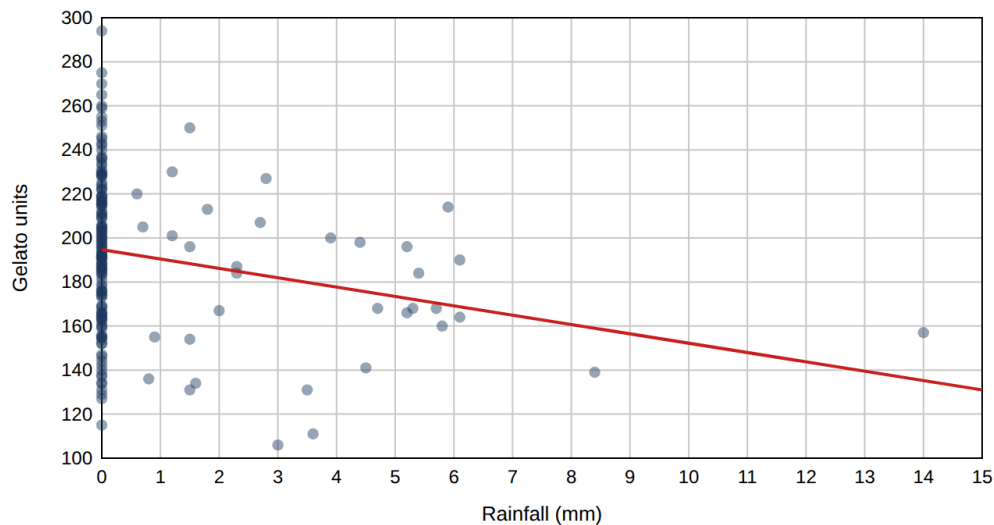
Stop. Finish Stage 1 before you turn the page. Do not change your Stage 1 answers after you see the data.

STAGE 2 · Check each variable on its own

≈ 7 min · 7 pts

Table 2. Summary of all 168 days (Sheets: AVERAGE, MEDIAN, STDEV.S, MIN, MAX). Rainfall was exactly 0 mm on 133 of the 168 days.

Variable	Mean	Median	Std dev	Min	Max
temperature_c	22.80	22.90	3.43	12.6	31.8
sunshine_hours	7.17	7.35	2.48	1.0	13.0
rainfall_mm	0.76	0.00	1.92	0.00	14.00
wind_kmh	13.53	13.20	6.98	3.0	26.0
humidity_pct	66.58	67.00	8.56	45.0	95.0
gelato_price_eur	4.27	4.30	0.39	3.70	4.80
gelato_units	191.44	192.00	35.84	106.00	294.00

Figure 1. Gelato units against rainfall for each of the 168 days, with the trendline Sheets draws. Many dots sit on top of each other at 0 mm.

- 5 Shape and extremes.** (a) Calculate the z-score of the **wettest day** (rainfall_mm = 14.0) and of the **hottest day** (temperature_c = 31.8). (b) Use the mean and median in Table 2 to say whether rainfall_mm and temperature_c are **symmetric or skewed**, and in which direction. 2 pts

- 6 A weak predictor?** Your intuition says rain should reduce gelato sales, yet r between rainfall_mm and gelato_units is only -0.23 ($R^2 = 0.05$). When rainfall_mm is added to Model 4 (Table 4), its t-statistic is -4.1 . When the **11 days with 5 mm or more** are removed ($n = 157$), it becomes -1.6 . (a) Use Table 2 and Figure 1 to explain why rainfall_mm, measured in millimetres, is a weak predictor on most days. (b) What does the change in the t-statistic tell you about where the evidence for a rain effect comes from? (c) rainfall_mm is in column E of the sheet. Write the Sheets formula for row 2 that derives **rainy_day** (1 if there was any rain, 0 otherwise). 3 pts

- 7 An extreme outcome.** The highest gelato_units is **294**, on Aug 2: temperature_c 30.9, weekend, beach_event = 1, price €4.80. (a) Calculate its z-score. (b) A classmate says: "Delete it, it is an outlier." Using the facts below, argue whether to delete it. **Facts:** Model 4 predicts **279** units for that day, and the SE of y for Model 4 is 19.6. Refitting the temperature-only model without that day changes its slope from 6.05 to 5.82. 2 pts

STAGE 3 · Check how the variables relate to each other

≈ 8 min · 7 pts

Table 3. Correlations (Sheets: CORREL) between gelato_units and the candidate predictors, all 168 days. Each cell is r for the row variable and the column variable (the table is symmetric, so only the lower half is shown).

	units	temp	sun	rain	wind	humid	event	wkend	price
temperature_c	0.58	1							
sunshine_hours	0.49	0.78	1						
rainfall_mm	-0.23	-0.15	-0.33	1					
wind_kmh	-0.01	-0.06	-0.01	0.00	1				
humidity_pct	-0.36	-0.46	-0.42	0.47	0.10	1			
beach_event	0.40	0.19	0.09	-0.13	0.16	-0.16	1		
weekend	0.42	-0.02	-0.02	0.10	-0.08	0.05	0.01	1	
gelato_price_eur	0.02	0.45	0.32	-0.12	0.05	-0.21	0.21	-0.03	1
post_change	0.26	0.56	0.40	-0.17	0.07	-0.25	0.30	0.00	0.76

- 8 Find the patterns.** (a) Which one predictor has the strongest linear relationship with gelato_units? Give its r. (b) List **every pair of predictors** with $|r| > 0.7$ (ignore the sign).

2 pts

- 9 Overlapping predictors.** Take the pair of **weather** variables you found in 8(b). (a) Calculate r^2 for the pair and say what it means. (b) Explain why putting both in the same LINEST model is a problem. (c) For **each** pair in 8(b), say which variable you would keep, using both Table 3 and common sense.

3 pts

- 10 The simplest model, by hand.** Use Tables 2 and 3 to build the simple regression of gelato_units on temperature_c: calculate the slope $b = r \times s_y \div s_x$ and the intercept $a = \bar{y} - b \times \bar{x}$, then predict gelato_units for a **28 °C** day.

2 pts

STAGE 4 · Build and compare models with LINEST

≈ 14 min · 13 pts

Table 4. Five models of gelato_units fitted with LINEST(..., TRUE, TRUE) to all 168 days (k = number of predictors).

Model	Predictors	k	R ²	SE of y
M1	temperature_c	1	0.335	29.31
M2	temperature_c, sunshine_hours	2	0.339	29.32
M3	temperature_c, weekend, beach_event	3	0.612	22.52
M4	temperature_c, weekend, beach_event, gelato_price_eur	4	0.707	19.63
M5	temperature_c, sunshine_hours, rainfall_mm, wind_kmh, humidity_pct, beach_event, weekend, gelato_price_eur	8	0.737	18.83

Table 5. Coefficients (standard errors) for two predictors in Models 1 and 2.

Model	temperature_c	sunshine_hours
M1	6.05 (0.66)	not in model
M2	5.28 (1.06)	1.36 (1.47)

- 11 Does adding sunshine help?** (a) Calculate the t-statistic for temperature_c in M1 and in M2, and for sunshine_hours in M2. (b) Calculate the adjusted R² for M1 and M2 (n = 168). (c) Use Table 3 to explain what happened to the temperature_c coefficient and its standard error when sunshine_hours was added.

3 pts

Table 6. The raw LINEST output for Model 4 (gelato_units on temperature_c, weekend, beach_event, gelato_price_eur; X columns in that order), all 168 days.

	Col 1	Col 2	Col 3	Col 4	Col 5
Row 1	-32.47	29.83	33.47	7.12	151.66
Row 2	4.47	3.76	3.35	0.50	17.50
Row 3	0.71	19.63	#N/A	#N/A	#N/A
Row 4	98.50	163.00	#N/A	#N/A	#N/A
Row 5	151763.41	62785.99	#N/A	#N/A	#N/A

- 12 Decode the LINEST output.** (a) Match each number in Row 1 to its variable (or the intercept). (b) Calculate the t-statistic for the price variable and interpret its coefficient in one sentence, **holding the other variables constant**. (c) Look back at your Stage 1 answers: name one expectation that LINEST supports, and one that it does not support (or one variable you left out that LINEST says matters).

3 pts

- 13 A price paradox.** The café raised the gelato price on 30 June. Daily gelato units: **before** (n = 84) mean 182.1, sd 35.3; **after** (n = 84) mean 200.7, sd 34.1. (a) Calculate the SE of the difference and t; is the difference statistically significant? (b) The owner concludes “higher prices raise sales”, yet the price coefficient in Model 4 is -32.47 (t = -7.3). Explain how both can be true. (c) Table 3 shows r ≈ 0.02 between price and gelato_units. Why can a variable that matters have r near 0 with the outcome?

3 pts

- 14 Predict, then check the residual.** Use Model 4 (Table 6) to predict gelato_units for each day below and calculate the **residual** (actual – predicted). Call a day **unusual** if $|\text{residual}| > 2 \times \text{SE of } y$ (19.6). Then say what you would do about an unusual day. 2 pts

Day	Model 4 inputs	Actual	Predicted
May 21	temperature_c = 24, weekend = 0, beach_event = 0, gelato_price_eur = 4.3	186	
Sep 4	temperature_c = 25, weekend = 0, beach_event = 0, gelato_price_eur = 4.5	127	

- 15 Let the residuals guide you.** Figure 2 plots the residuals of Model 1 (temperature only). Table 7 gives their mean by group. (a) Which **one** omitted variable should you add first, with what sign, and how can you tell from the figure and the table? (b) On the empty axes, sketch the residual plot you would hope to see for the improved model. 2 pts

Figure 2. Residuals of Model 1 against temperature, one dot per day: red filled dots are weekend days, hollow dots are weekdays. The dashed line is zero.

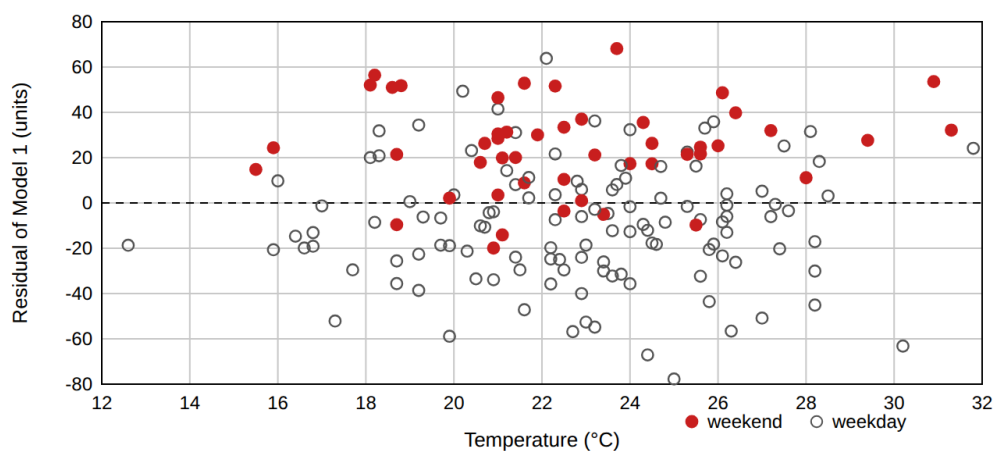
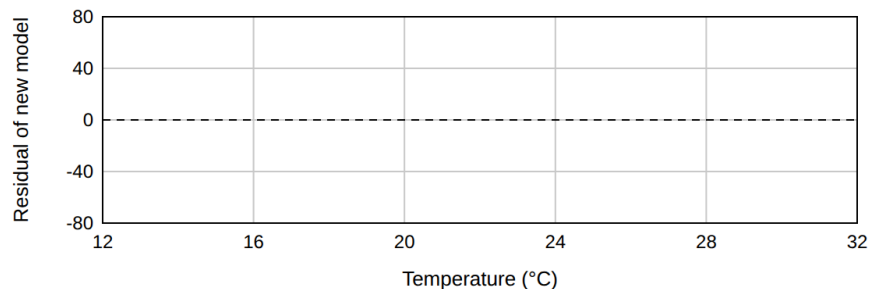


Table 7. Mean residual of Model 1 by group (all 168 days).

Group	Days	Mean residual (units)
weekend = 1	48	+24.7
weekend = 0	120	–9.9
beach_event = 1	37	+19.9
beach_event = 0	131	–5.6
price ≤ €4.00	57	+9.2
price between	49	–0.3
price ≥ €4.50	62	–8.2

Space for your sketch in 15(b):



STAGE 5 · Does the model forecast unseen days?

≈ 8 min · 7 pts

Models 4 and 5 were **refitted using only the first 112 days** (7 April–27 July). Their R^2 on those 112 days is **0.708** (M4) and **0.752** (M5). Table 8 shows 4 of the 56 later days that neither model has seen. A third forecaster, the **baseline**, uses no model: it takes a **weighted average of the last three same-weekday sales** with weights **0.5** (one week ago), **0.3** (two weeks ago) and **0.2** (three weeks ago).

Table 8. Four unseen days.

Day	Actual	1 week ago	2 weeks ago	3 weeks ago	M4 predicts	M5 predicts
Aug 3	205	275	201	233	204.8	208.6
Aug 8	219	166	243	199	202.5	200.4
Aug 18	140	224	201	191	161.2	161.5
Aug 19	212	195	188	246	192.0	195.1

Table 9. Space for your errors.

- 16 Compute the forecast errors.** (a) Calculate the baseline forecast for each day. (b) Complete Table 9 with the absolute error of each forecaster on each day, and the **MAE** of each. (1 mark for the baseline forecasts, 2 for the errors, 2 for the three MAEs.)

5 pts

Day	Baseline forecast	error baseline	error M4	error M5
Aug 3				
Aug 8				
Aug 18				
Aug 19				
MAE	—			

- 17 In-sample fit versus forecasting.** M5 has the higher R^2 on the 112 training days. (a) Explain why that does not guarantee that M5 forecasts unseen days better. (b) Which model would you give the owner, and why? Think about what the café would have to know the evening before.

2 pts

STAGE 6 · Advise the owner and extend the model

≈ 8 min · 7 pts

- 18 Advice for the owner.** Write 3–4 sentences to the owner, who has no statistics background. Say which variables you recommend using to forecast gelato units, how far off a typical forecast will be (use a number from this exam, in cups), and one thing the model cannot tell the owner.

3 pts

- 19 Your own idea.** The owner can start recording **one new column** of data tomorrow. Choose a variable that is **not** in the table on page 1. Tell me: (i) what it is and exactly how it would be measured; (ii) the sign you expect and why; (iii) how you would use LINEST and unseen days to decide whether it earns a place in the model; (iv) a result that would show you were wrong. Your reasoning matters more than being right.

4 pts

Reference sheet (you may use this on any question)**z-score** = $(\text{value} - \text{mean}) \div s$ · treat $|z| > 2$ as unusual**t-statistic** = coefficient $\div$ its standard error · treat $|t| > 2$ as statistically significant (5% level) **r^2** = share of variation shared by two variables · two predictors with $|r| > 0.7$ overlap heavily**Simple regression from summaries:** slope $b = r \times s_y \div s_x$ · intercept $a = \bar{y} - b \times \bar{x}$ · prediction $\hat{y} = a + b \times x$ **Residual** = actual – predicted · **MAE** (mean absolute error) = average of the absolute residuals**Adjusted R^2** = $1 - (1 - R^2)(n - 1) \div (n - k - 1)$, n = number of days, k = number of predictors**Difference of two means:** $SE = \sqrt{(s_1^2/n_1 + s_2^2/n_2)}$, $t = (\text{mean}_2 - \text{mean}_1) \div SE$ **Weighted average** = $\Sigma(\text{weight} \times \text{value}) \div \Sigma(\text{weights})$ **LINEST(y, X, TRUE, TRUE)** with k predictor columns returns 5 rows: Row 1 coefficients **in reverse order** (last X column first, ... first X column, intercept last) · Row 2 their standard errors · Row 3: R^2 , SE of y (typical size of a residual) · Row 4: F , $df (= n - k - 1)$ · Row 5: SS regression, SS residual (#N/A fills the empty cells)**Scratch space**

Write the question number beside anything you want marked.

— End of exam —